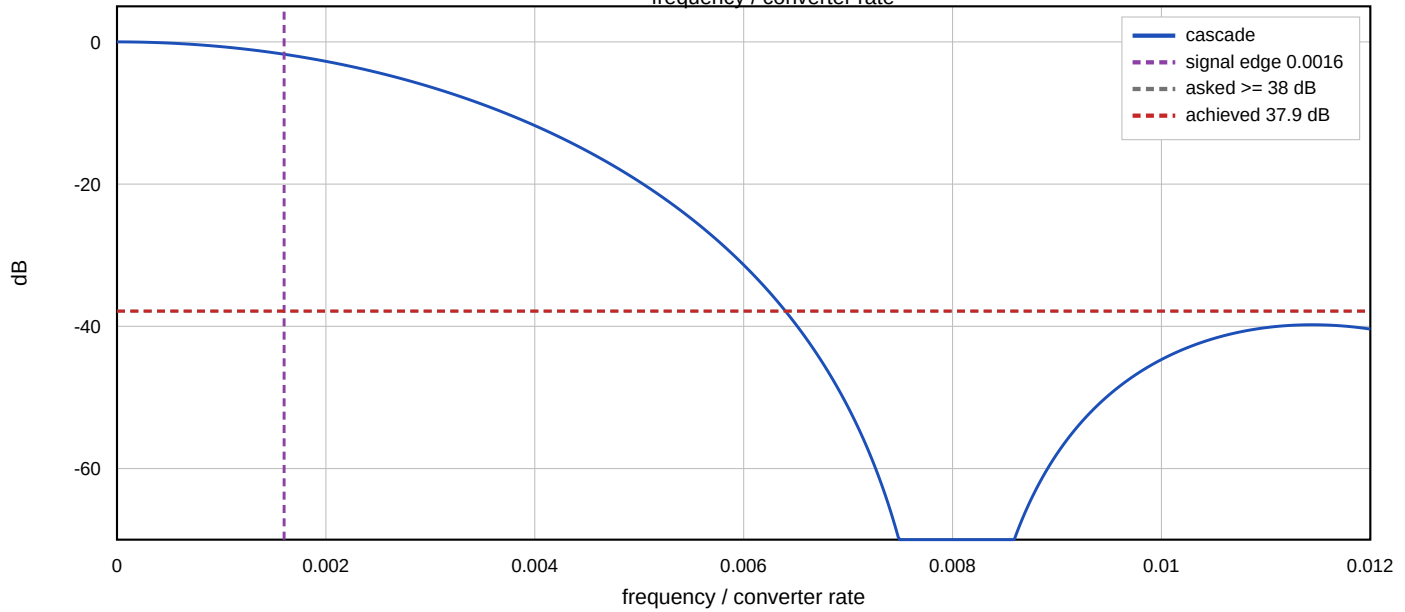
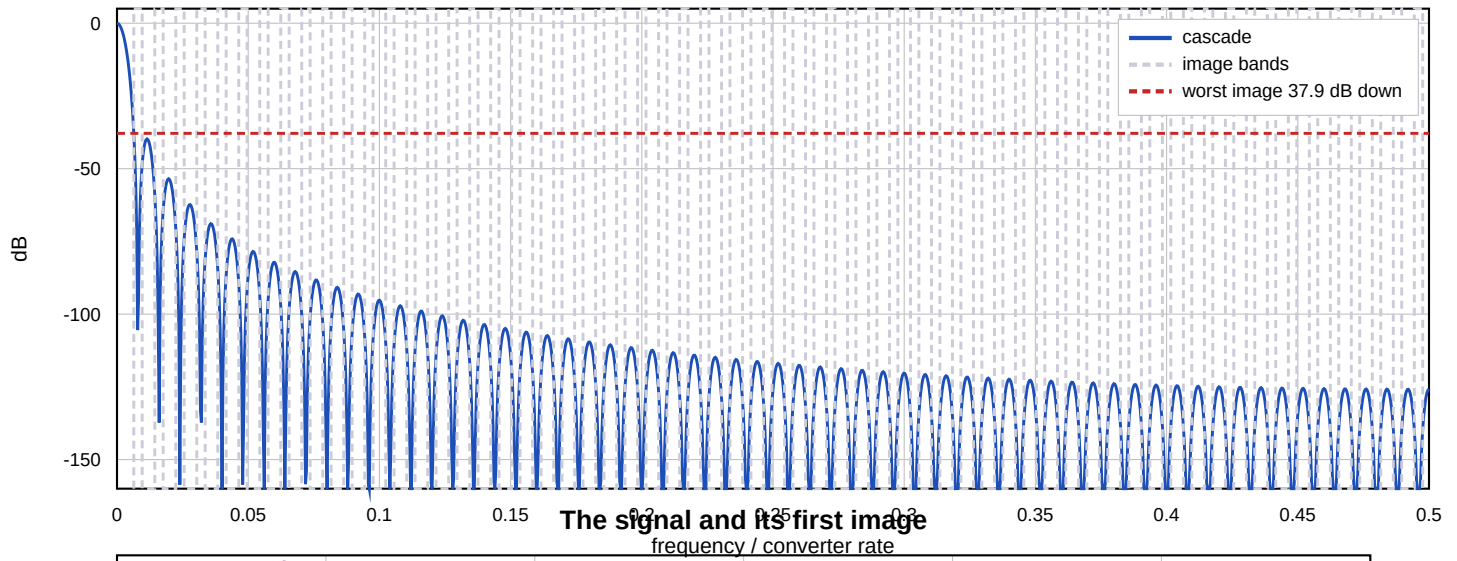


Interpolation Chain - Specified Parameters

1000.0000 kHz x 125 = 125.000 MHz, split [125], 200.000 kHz signal

Composite magnitude across the converter band



Stages

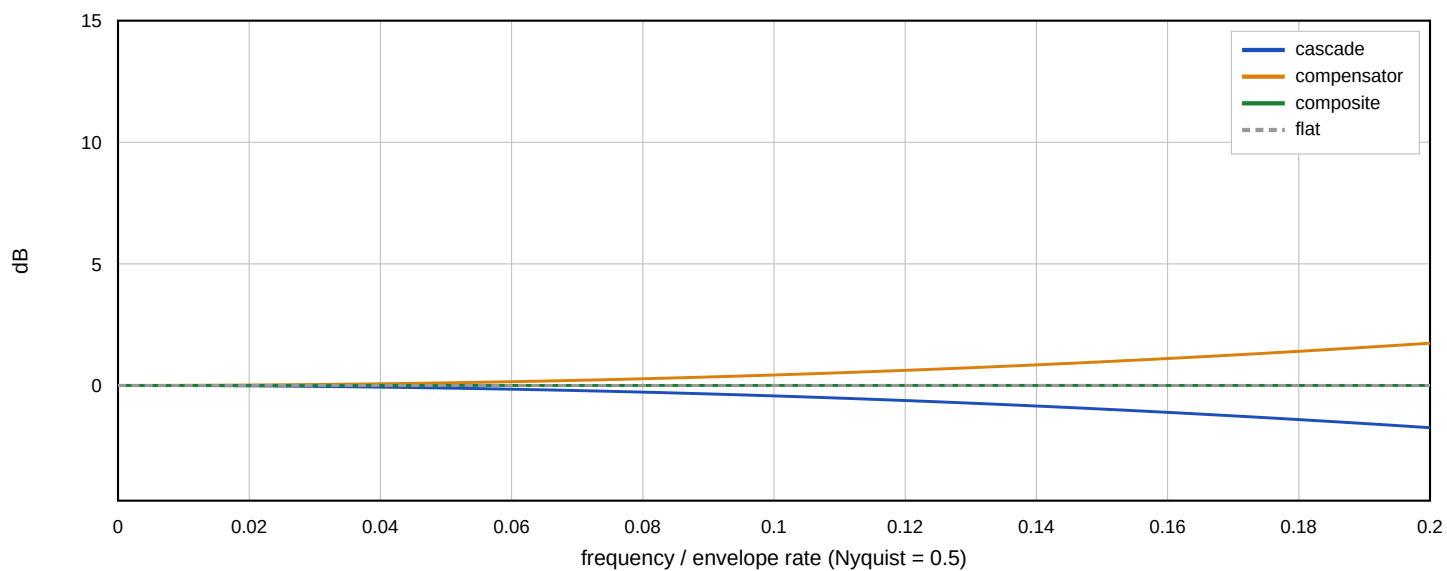
stage 1: x125 N=3 M=1 combs at 1.0000 MHz, integrators at 125.000 MHz
in 16 bits, uniform accumulator 30 bits, tapered widths [17, 18, 19, 18, 24, 30]
270 register bits uniform, 181 as built (lossless), bound 180
widths as built [17, 18, 19, 19, 24, 30]

The combs run at the stage's input rate and the integrators at its output rate, so an integrator bit costs R times what a comb bit costs. Depth belongs early and rate late. The tapered widths are lossless: each stage sized to its own growth bound holds its value exactly, so a tapered interpolator is bit-identical to a uniform one.

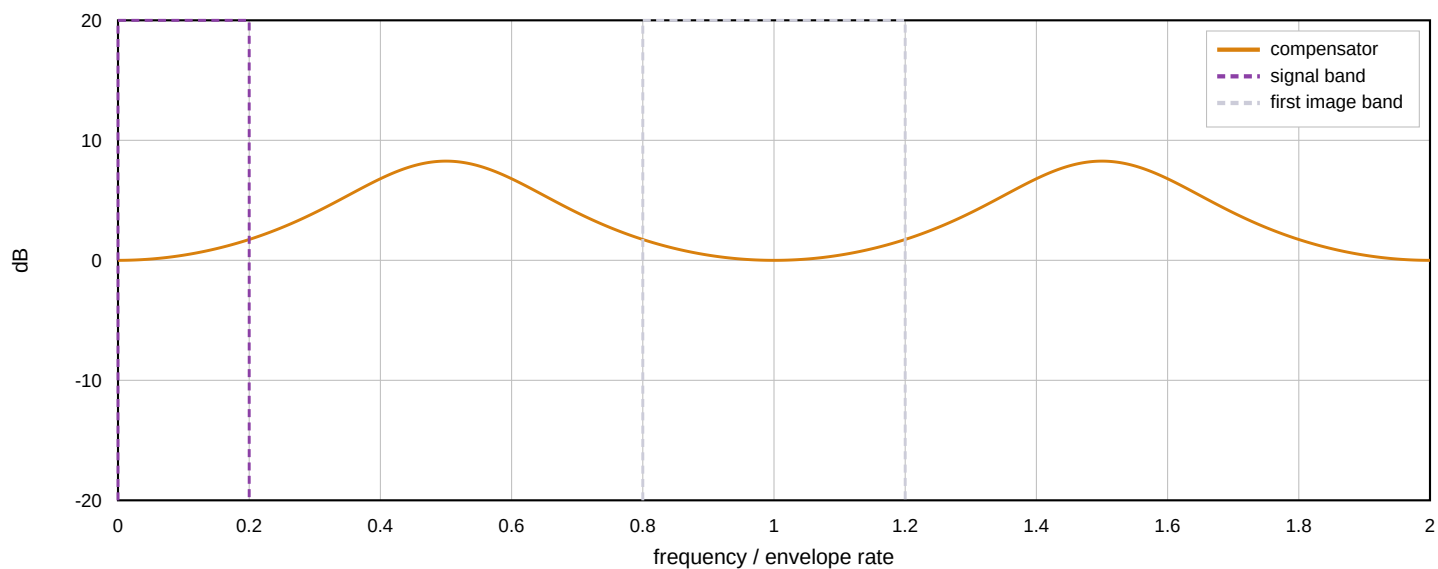
Compensation and Result

15 taps, 16-bit coefficients, at the 1.0000 MHz envelope rate

Droop, pre-compensator, and the two together - the signal band



The compensator repeats, so it lifts every image as much as the signal



Figures and Cautions

Figures

images 37.9 dB down (asked ≥ 37.9)
 worst image at 0.8000 MHz
passband ripple 0.0011 dB (asked ≤ 0.001)
uncompensated droop .. -1.738 dB at the band edge
converter floor 86.0 dB at 14 bits (the chain adds none)
register bits 270 uniform, 181 as built, 180 at the exact bound
compensator DC gain .. 1.0000, shift 14
compensator headroom . 18 bits for any input, 17 for in-band only
 norms I1 2.5906, passband peak 1.2214
rates reachable 124 of 124 in 2..=125 (single stage: all of them)
worst image at rate .. 125
adder depth 1 in the combs, 1 in the integrators (both pipelined)

group delay 1439 converter samples = 11.5 μ s
 cascade 186, integrator pipeline 2, comb pipeline 375, output regs 1, compensator 875
 largest term is the compensator at 875 (61%); loop bandwidth ~ 8.7 kHz

taps [0, 4, -27, 130, -506, 1737, -5982, 25672, -5982, 1737, -506, 130, -27, 4, 0]

Build to the any-input headroom unless the envelope is band-limited.

The in-band figure is the peak gain an in-band sinusoid sees. The any-input figure is the sum of the taps' magnitudes, which is the bound for an arbitrary bounded input -- and a transmit envelope is not band-limited: switch-on, a burst boundary and a modulation change are all steps, and a step is exactly the input that reaches it. Choosing the narrower figure for a burst transmitter is how a compensator saturates on the first sample.

More taps will not improve image rejection.

The compensator runs before the rate change, at the envelope rate, so its response is periodic and the image at $k+u$ sees exactly the gain the signal at u sees -- the plot above shows it. The image-to-signal ratio is the cascade's alone. This is the one place where transposing receive-side intuition is wrong: there, the compensator sits after the fold and its stopband is part of the alias budget.

The knobs are the CIC depth N , the interpolation factor R , and the signal bandwidth. A compensator at the converter rate would work, because it is no longer periodic in u , and costs a FIR running at the full clock.